



I run **aifutures.dev**, an independent product lab building tools for the agentic age. The software there trades live markets, maps living codebases and runs fleets of autonomous agents, always with a person in the loop where it counts. Full-stack developer with an M.S. in Computer Science. Before this I shipped production marketplace apps used by 200+ people at Airbnb. How I work: machine speed, human judgment, provable evidence.

EXPERIENCE

Founder & Software Engineer, tpochart.tech

2026 to now

Futures Trading Platform · Chicago, IL · [tpochart.tech](#)

- Built and operate a production futures trading platform end to end using TypeScript, Node.js, Next.js, React, and PostgreSQL through Supabase
- Integrated ProjectX and TopstepX REST and SignalR APIs for live market data, account state, and order events, with reconnect, polling fallback, token refresh, rate-limit handling, and contract-roll safeguards
- Designed a distributed architecture that keeps broker credentials and trade execution on a local worker while the cloud dashboard coordinates signals, confirmations, and durable state
- Implemented fail-closed risk controls including confirmation-time validation, server-side stop and target brackets, execution leases, stale-feed gates, reconciliation, and a position-flattening kill switch
- Own features from design through deployment with automated tests, database migrations, health checks, operational logging, incident runbooks, and production monitoring

Computer Information Systems Tutor

Oct 2025 to now

Wilbur Wright College · City Colleges of Chicago · Chicago, IL

- Provide individual and group tutoring across CIS subjects: programming, databases, algorithms, and web development
- Lead workshops on study skills and applied problem-solving; develop personalized learning plans aligned with course objectives

JavaScript Developer I

Feb 2023 to Nov 2023

AXANEXA · Chicago, IL

- Architected and delivered a resource management application on monday.com for Airbnb, adopted by 200+ team members for day-to-day project coordination and resource allocation
- Launched the company's first Marketplace listing, a Dependent Dropdown app now used across several client organizations
- Built a Role Assignment system capable of processing hundreds of concurrent tasks with intelligent API rate-limit queuing and GraphQL batch optimization
- Pioneered reactive UI/UX design patterns across desktop and mobile, reducing user onboarding friction and increasing adoption

Teaching Assistant, Accelerated Intro to CS

Aug 2016 to Dec 2017

Illinois Institute of Technology · Chicago, IL

- Assisted professor in lab sessions for 60+ students; held weekly office hours and graded assignments with detailed feedback
- Identified common misconceptions and developed targeted exercises that improved class average by ~12%

Digital Communications, Institutional Advancement

Nov 2015 to Dec 2017

Illinois Institute of Technology · Chicago, IL

- Administered IIT Alumni and Mies Van der Rohe Society websites; migrated legacy media database improving load performance by 40%
- Produced Giving Day 2016 promotional video; published digital content reaching 10,000+ alumni and donors

Mathematics Tutor

Jun 2014 to May 2015

Wilbur Wright College · City Colleges of Chicago · Chicago, IL

- Delivered personalized math tutoring (Algebra through Calculus) to 30+ students per semester with 95%+ satisfaction rating

OTHER PROJECTS

ClawSwarm · Multi-Agent AI Orchestration (OSS + hosted platform)

Autonomous orchestration with three specialized chiefs: CodeClaw (code), ResearchClaw (research), OpsClaw (infrastructure). 3-tier quality gate: ≥8 auto-approves, 5–7 escalates to human review, <5 auto-reworks. Real-time WebSocket streaming dashboard, replay logging, per-agent workspace isolation. ~65% auto-approve across 50+ soak tests. [clawswarm.app](#)

TypeScript · Convex · Next.js · WebSockets · AI Architecture

AgentAwake · AI Productivity SaaS + Agent Memory Playbook

Full-stack SaaS with tiered Stripe billing, automated Resend drip sequences and an audience pipeline, all confirmed end to end in production. Includes a free 36-chapter playbook on building AI agents with persistent memory (Claude, ChatGPT, CrewAI, LangChain, n8n & more). [agentawake.com](#)

Next.js · TypeScript · Stripe · Resend · Vercel · SaaS

Market Profile Guide & ETH Smart Contracts

Interactive trading education platform for market profile theory and price-action structure ([live on Vercel](#)) · Ethereum smart contract development platform streamlining write–test–deploy workflows for on-chain apps.

Next.js · React · Tailwind CSS · Solidity · Web3.js

EDUCATION

M.S. Computer Science

California State University, Fullerton

B.S. Computer Science

Illinois Institute of Technology

A.S. Computer Science

Wilbur Wright College

TECHNICAL SKILLS

LANGUAGES

JavaScript TypeScript Python Solidity

FRONTEND

React Next.js Tailwind CSS
Framer Motion

BACKEND & APIS

Node.js GraphQL REST APIs
WebSockets

PLATFORMS & TOOLS

Monday.com SDK Make.com Vercel
Stripe Convex Resend Git / GitHub

AI & ARCHITECTURE

Agent Systems LLM Integration
Multi-Agent Orchestration RAG / Retrieval
MCP AST Analysis Local-First Systems

MARKETS & SECURITY

Market Structure / TPO Risk Controls
Ed25519 / SHA-256

HONORS & AWARDS

- ✦ IIT Presidential Scholarship
- ✦ Class Salutatorian
- ✦ Phi Theta Kappa Honor Society
- ✦ All Illinois Academic Team
- ✦ Presidential Scholar Semi-finalist

LANGUAGES

🇻🇳 Vietnamese · Native

🇺🇸 English · Professional

INTERESTS

AI Systems Design · Agent Architecture
Education & Mentorship · Web3 / DeFi
Open Source · Market Structure Trading